ARIEL UNIVERSITY /Department of Computer Science

LowerLoss
FULL STACK MACHINE LEARNING PROJECT

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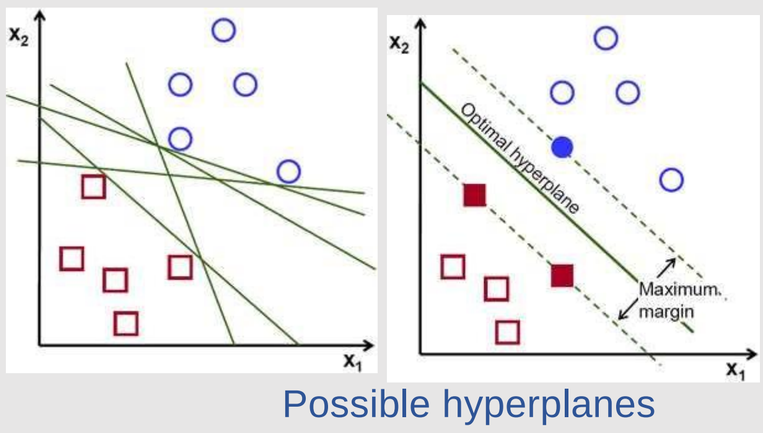
CONTRIBUTION

The purpose of the project is to give the customer a car price checking service with the forecast for a few years ahead. A customer gives details of the vehicle and in return receives a range of prices and percentages. The system conducts a survey of the car market in Israel and uses algorithm to calculate prices.
The algorithm Support Vector Machine (SVM) is a supervised machine learning algorithm classifier that represents the boundary that best separates the two classes (hyperplane / line). The hyperplane will help determine the average price among the data.
With the help of this service a customer will get the expected price of the vehicle is according to the market and not according to the official price list.
The price list:
1) Addresses young people.
2)The price list is based on statistics that are affected by the bid amounts of the vehicle owners. An algorithm will detect expensive and cheap bids and the intermediate bid that will be returned to the user.
3) The price list can predict prices for few years ahead.

ALGORITM SVM

What is Support Vector Machine?
The objective of the support vector machine algorithm is to find a hyperplane in an N-dimensional space(N — the number of features) that distinctly classifies the data points.



Possible hyperplanes

To separate the two classes of data points, there are many possible hyperplanes that could be chosen. Our objective is to find a plane that has the maximum margin, i.e the maximum distance between data points of both classes. Maximizing the margin distance provides some reinforcement so that future data points can be classified with more confidence. In our case, the client sends a request for a specific car make and model. The Back End processes this request and selects the necessary points. Each data point represents a specific make and model of a car, x - mileage y - price. The SVM algorithm separate the data into two classes.

INTRODUCTION

Today, when buying a new or used car in Israel, most buyers rely on the exterior and interior of the vehicle. After a few years, when they want to sell a car, they may be faced with a situation where the price they are asking for does not match the market price. The LowerLoss allows the buyer to see in advance their losses on the sale of the car. Also find out whether the amount that the buyer wants to spend on the car corresponds to the real price of the car. Perhaps some of the buyers in this regard will want to change their choice of car brand in order to reduce costs in the future.

ALTERNATIVES

The alternative to the SVM algorithm can be average price. But not clear between which points. Between the most expensive option and the cheapest? And if these points are very far from the accumulation of a larger number of points? And in this case, our initial data for calculating the price drop will be incorrect. Therefore, a support vector is preferable.

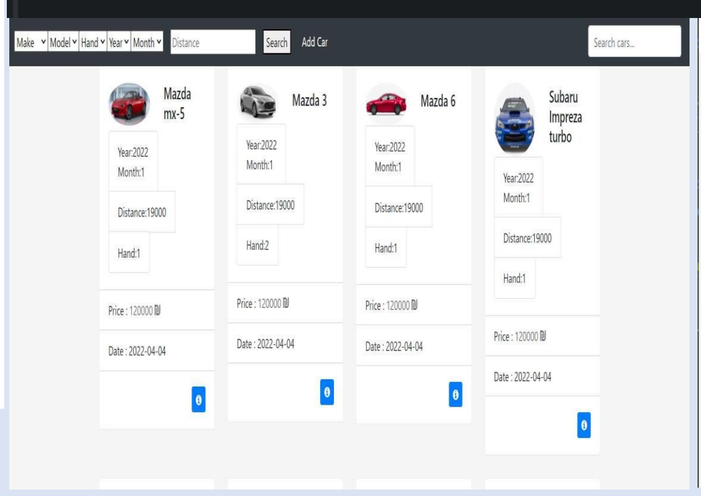
SELECTED APPROACH

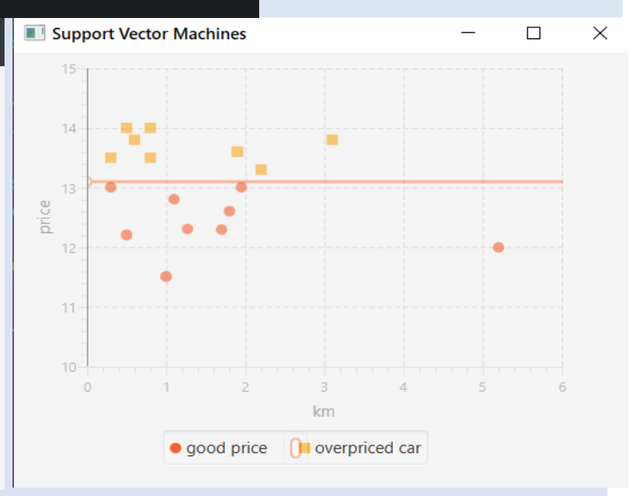
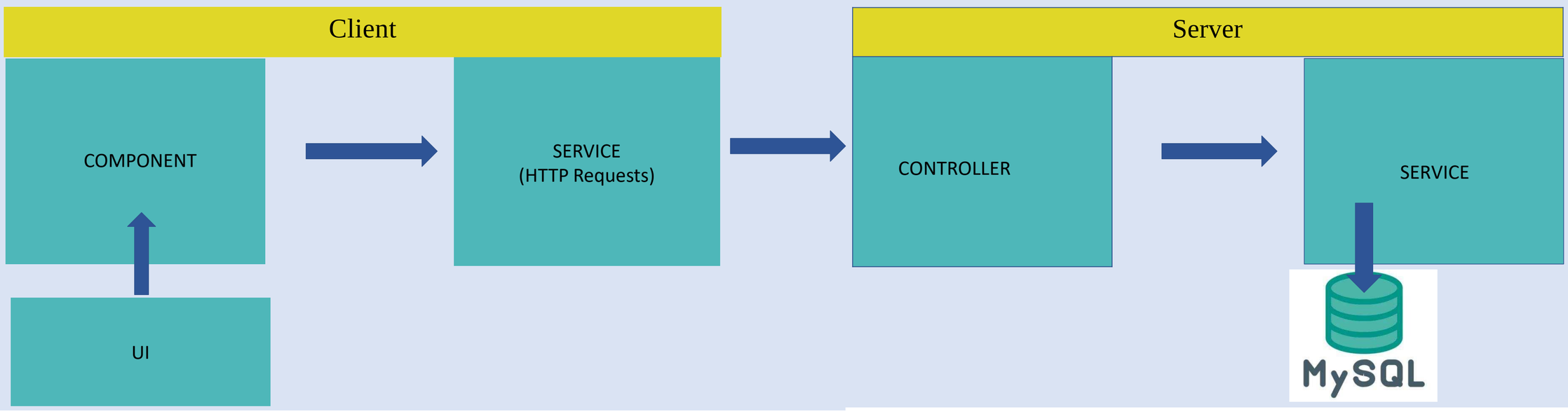
Support vectors are data points that are closer to the hyperplane and affect the position and orientation of the hyperplane. Using these support vectors to find the average car price from the existing data in the database and calculate the cost of the car 5 years in advance after the date of purchase. After that, points will be stored in data base. On every single query by API the program will turn to the database, which contains prepared information about the fall in prices. The program will calculate the price reduction and return to the client: information about the average maximum price for a particular car and the average minimum price from the existing ones in the database, a graph with all car options that are divided into two classes(High/Low price for now) and a price prediction table for few years ahead.

```
Support Vector      |   label | alpha
-----------------------------------------
[1.9, 13.6]|1.0   |0.2246272378
[1.8, 12.6]|-1.0  |0.5000972463
```

```
> to classify new car enter km and price 1 & 2 (or exit):
1.2 12.00
classify classified as -1 (very good car)

> to classify new car enter km and price 1 & 2 (or exit):
5.0 15.00
classify classification as 1 (not good car)
```





SOLUTION DESCRIPTION: MODULATION , UI